CCLAB-2

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Week-2

1. a) Create and attach an EBS volume to an EC2 instance, and
   b) scale the instance by increasing CPU and RAM by changing the instance type.

2.Attach and permanently mount an EBS volume to a Linux EC2 instance to ensure data persistence across reboots.

3. Create a snapshot of the attached EBS volume and use it to create and attach a new volume to an EC2 instance in another AWS region.

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1. a) Create and attach an EBS volume to an EC2 instance, and
   b) scale the instance by increasing CPU and RAM by changing the instance type.
   c) Creating and attaching an EBS volume to an EC2 instance

Step 1: Create a New EBS Volume

Open the EC2 Dashboard.

From the left-hand menu, select Volumes under Elastic Block Store (EBS).

Click Create Volume.

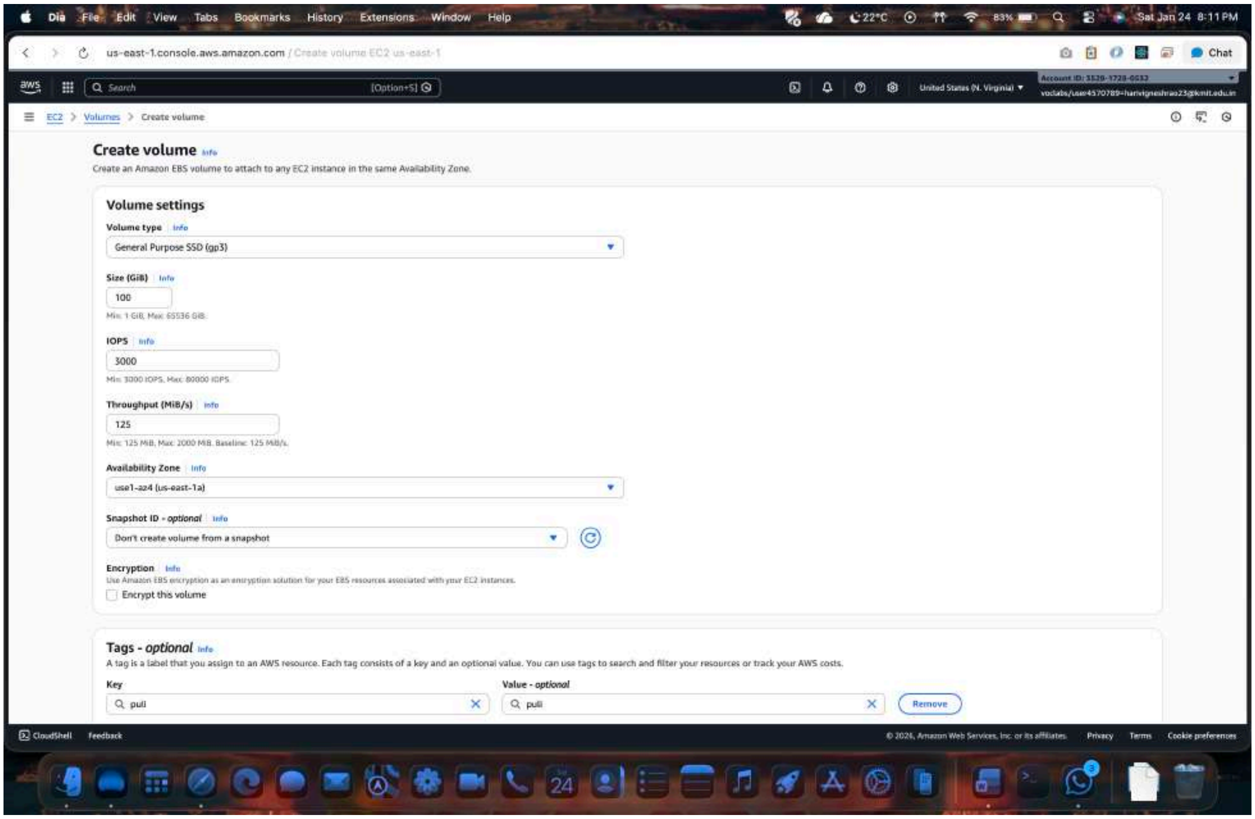
Choose:

Volume type (gp3/gp2)

Size of the volume

Availability Zone (must be the same as the EC2 instance)

Click Create Volume.



Step 2: Attach the EBS Volume to the Existing EC2 Instance

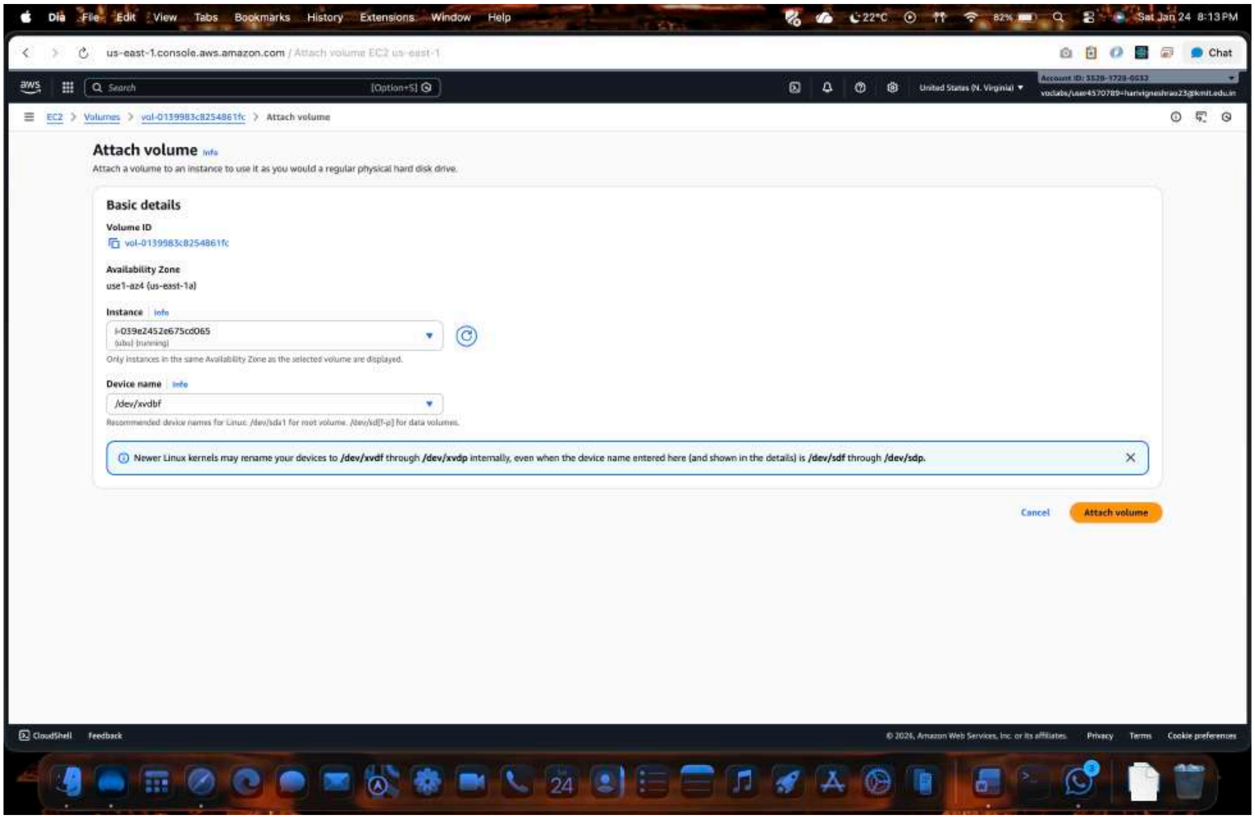
In the Volumes section, select the newly created volume.

Click Actions → Attach Volume.

Select the target EC2 instance from the list.

Specify the device name (example: /dev/xvdf).

Click Attach.



Step 3: Verify the Volume on the EC2 Instance

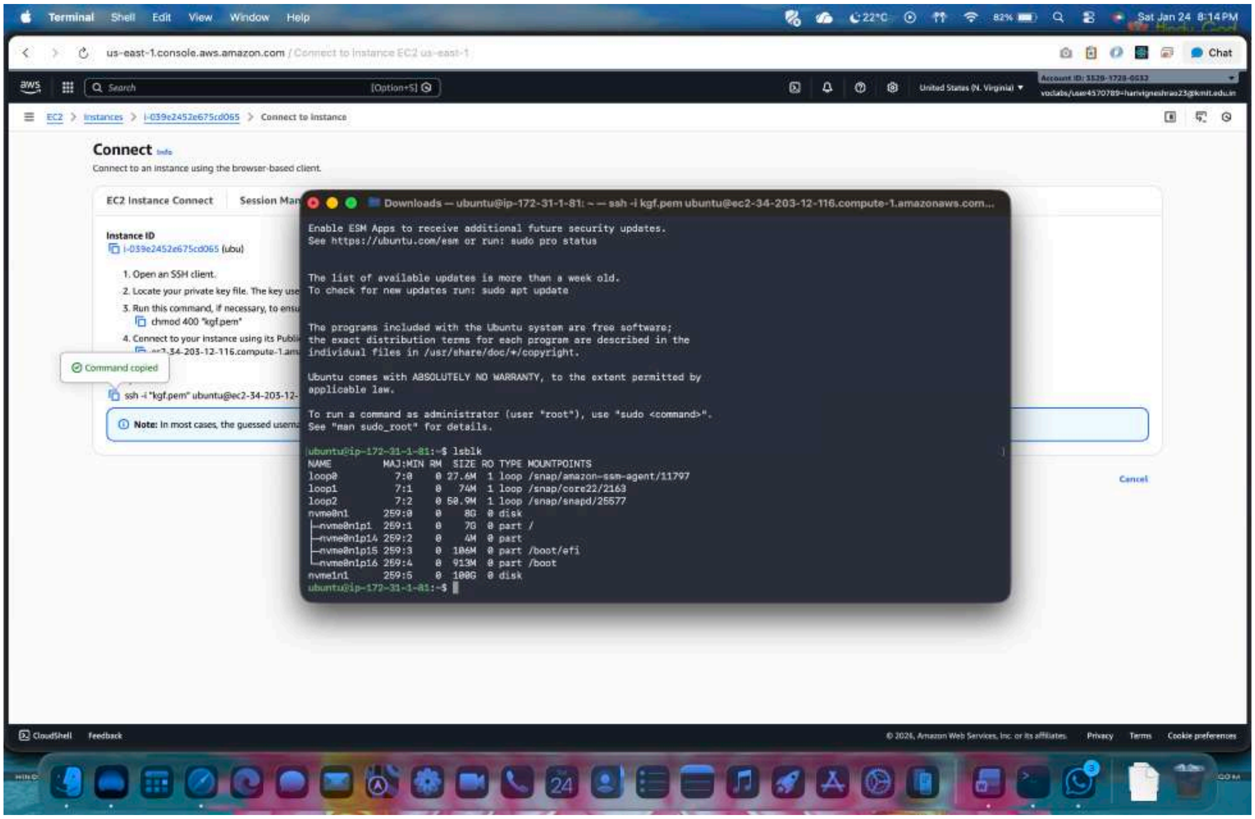
Connect to the EC2 instance using SSH.

Check the attached disks:

lsblk

Identify the new volume (e.g., /dev/xvdf).

Note: Ensure that the EBS volume and the EC2 instance belong to the **same Availability Zone**, as cross-AZ attachment is not supported.



b)Procedure: increasing/decreasing of CPU and RAM

Step 1. - Navigate to the EC2 dashboard.

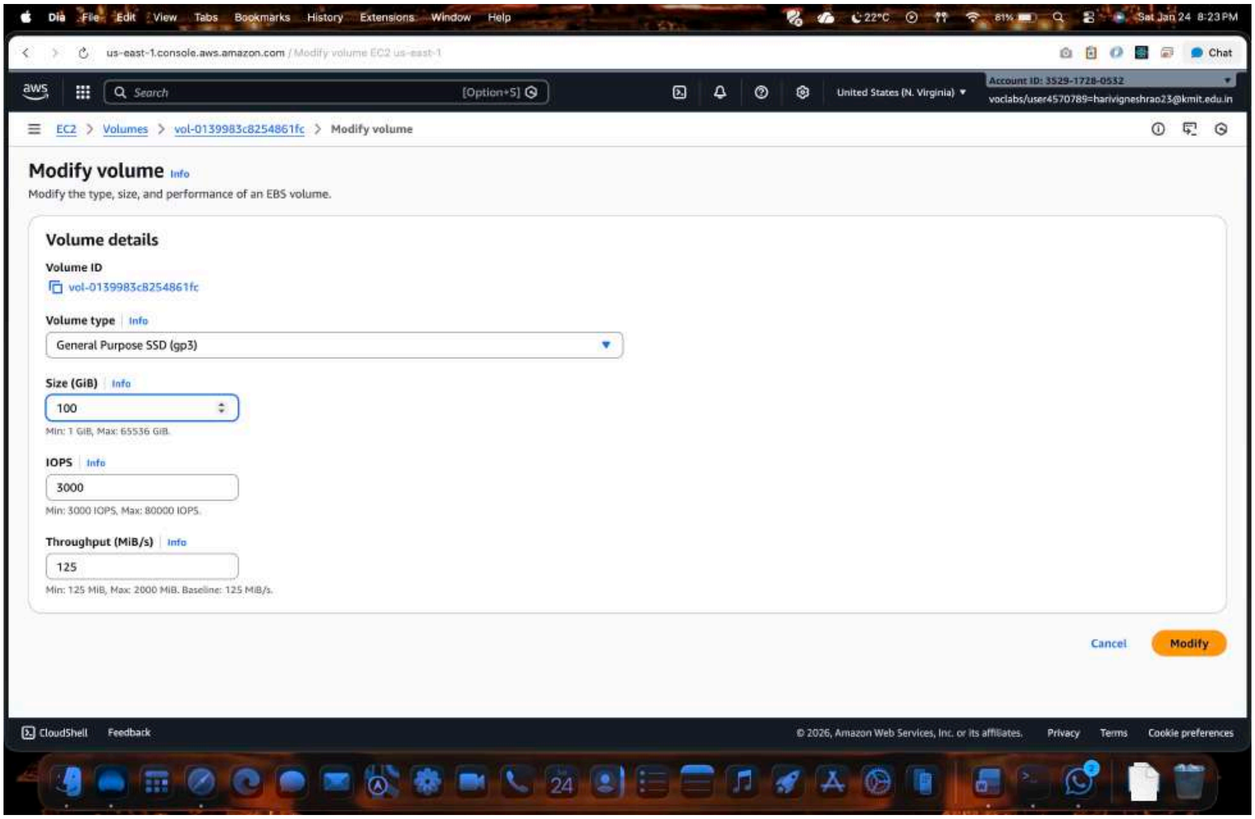
Step 2. - Stop the EC2 instance associated with the EBS volume.

Step 3. - Click "Actions" and then—instance settings—change the instance type"Modify Volume".

Step 4. -Increase/ decrease the size or change the volume type to a higher performance specification.

Step 5. - Click "Modify" to apply the changes.

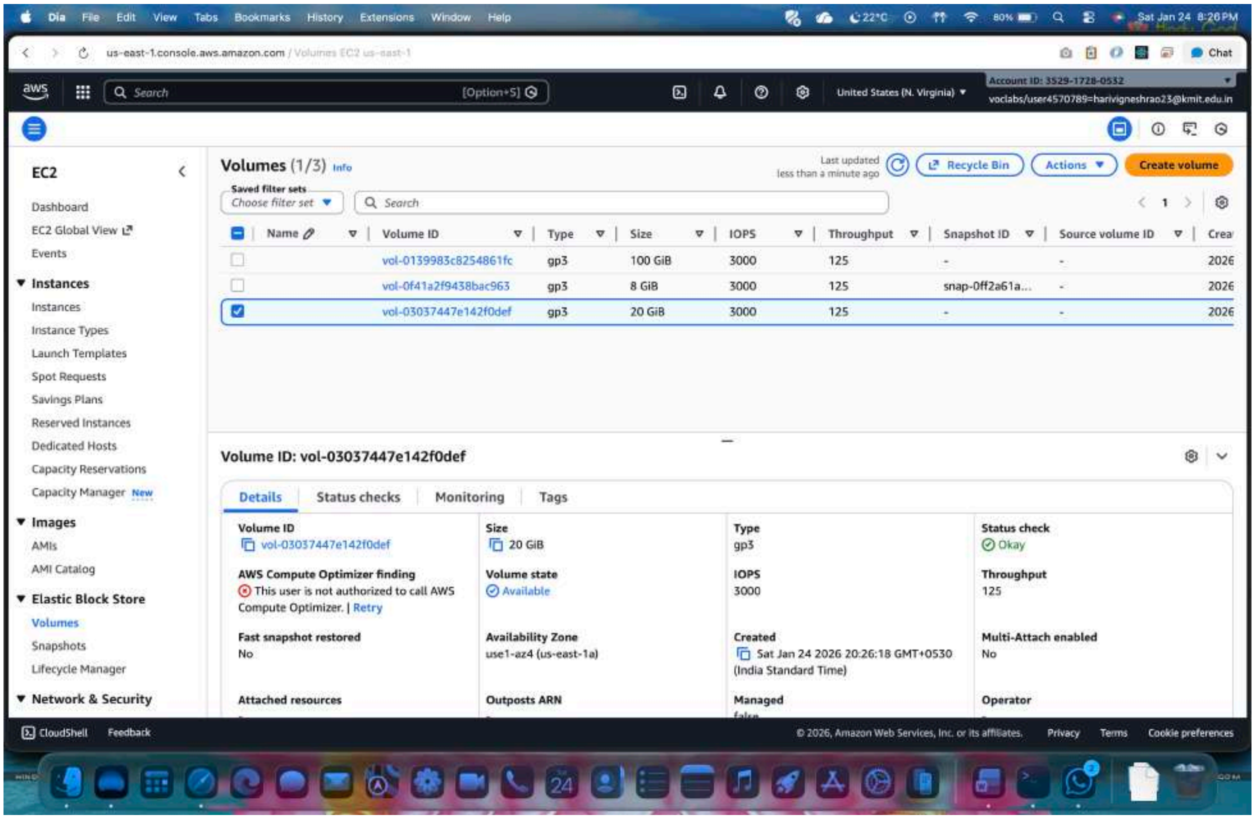
Step 6: Now the start the machine you see with increase /decrease of size and performance



2.Attach and permanently mount an EBS volume to a Linux EC2 instance to ensure data persistence across reboots.

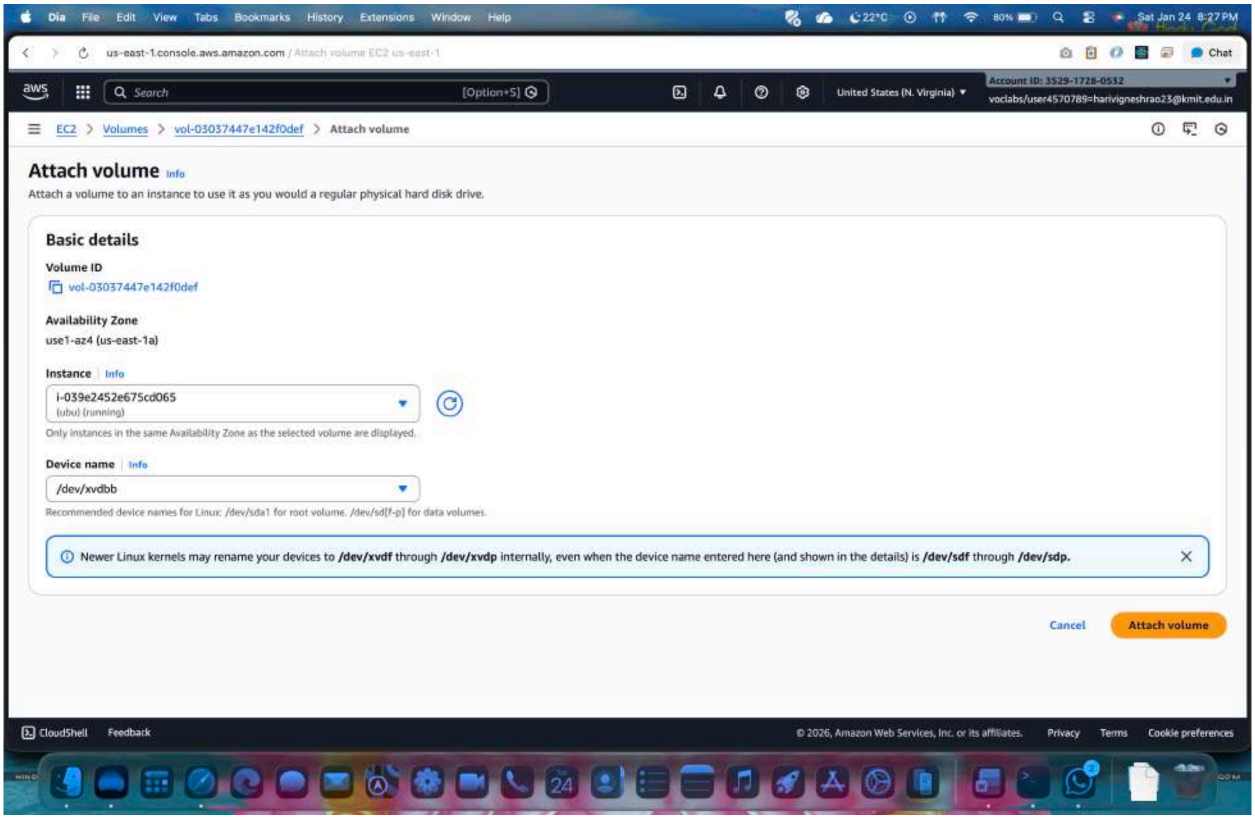
Step 1: Create an EBS Volume

1. Log in to **AWS Management Console**
2. Go to **EC2 → Volumes**
3. Click **Create Volume**
4. Choose:
   - **Volume Type**: gp3 (recommended)
   - **Size**: (e.g., 20 GB)
   - **Availability Zone**: **Same AZ as EC2 instance**
5. Click **Create Volume**



Step 2: Attach the EBS Volume to EC2

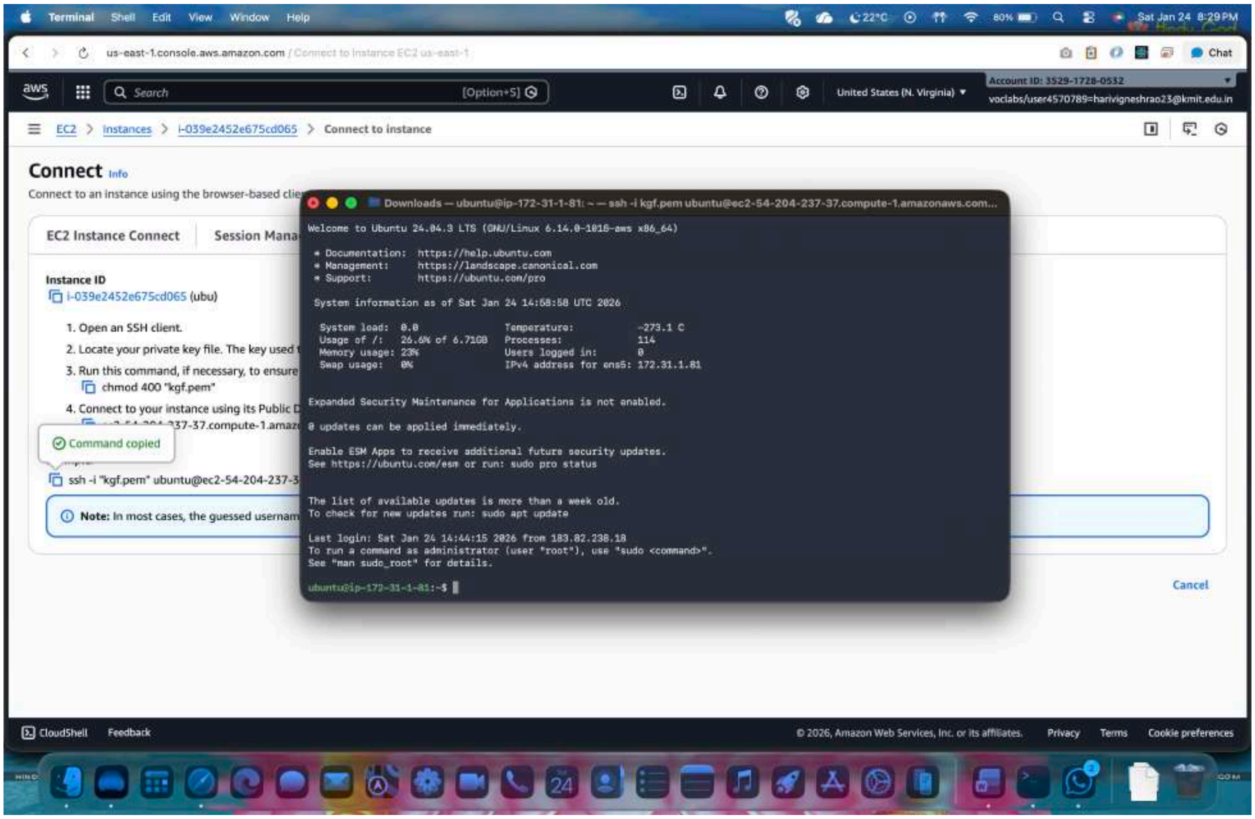
1. Select the newly created **EBS volume**
2. Click **Actions → Attach Volume**
3. Choose:
    - **Instance**: Select your Linux EC2 instance
    - **Device name**: /dev/xvdf
4. Click **Attach**



Step 3: Connect to the EC2 Instance

Connect using SSH:

ssh -i key.pem ec2-user@<public-ip>



Step 4: Verify the Attached Disk

Check all available disks:

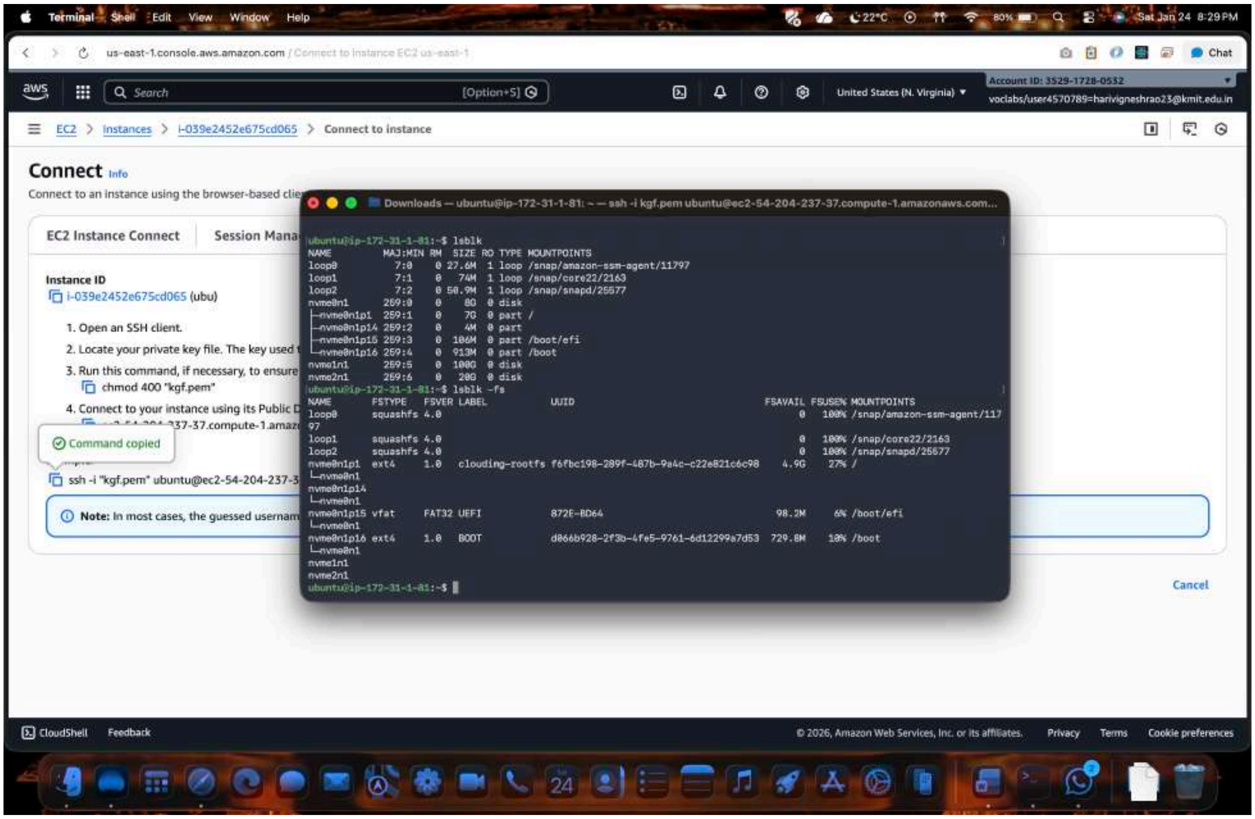
lsblk

You will see:

- Root volume: /dev/xvda
- New volume: /dev/xvdf (no partition yet)

Check filesystem:

lsblk -fs



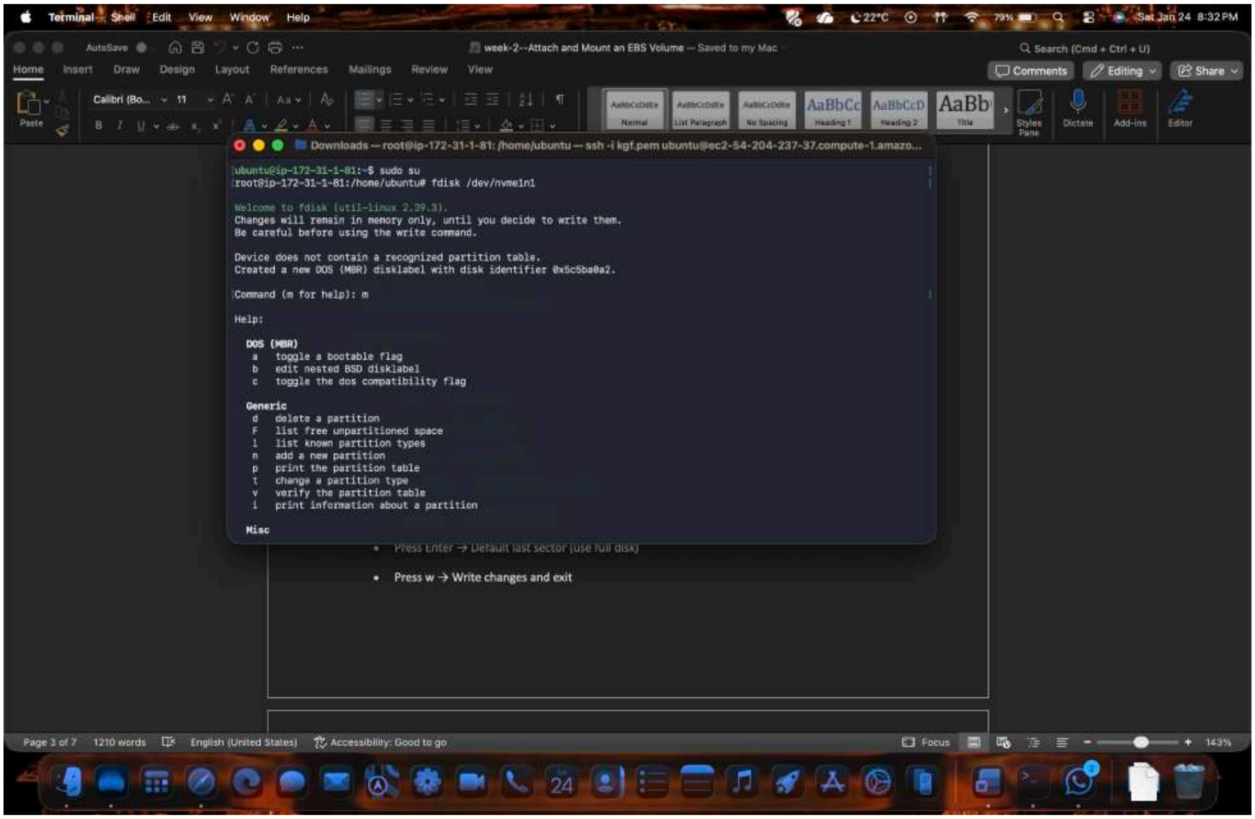
Step 5: Create a Partition

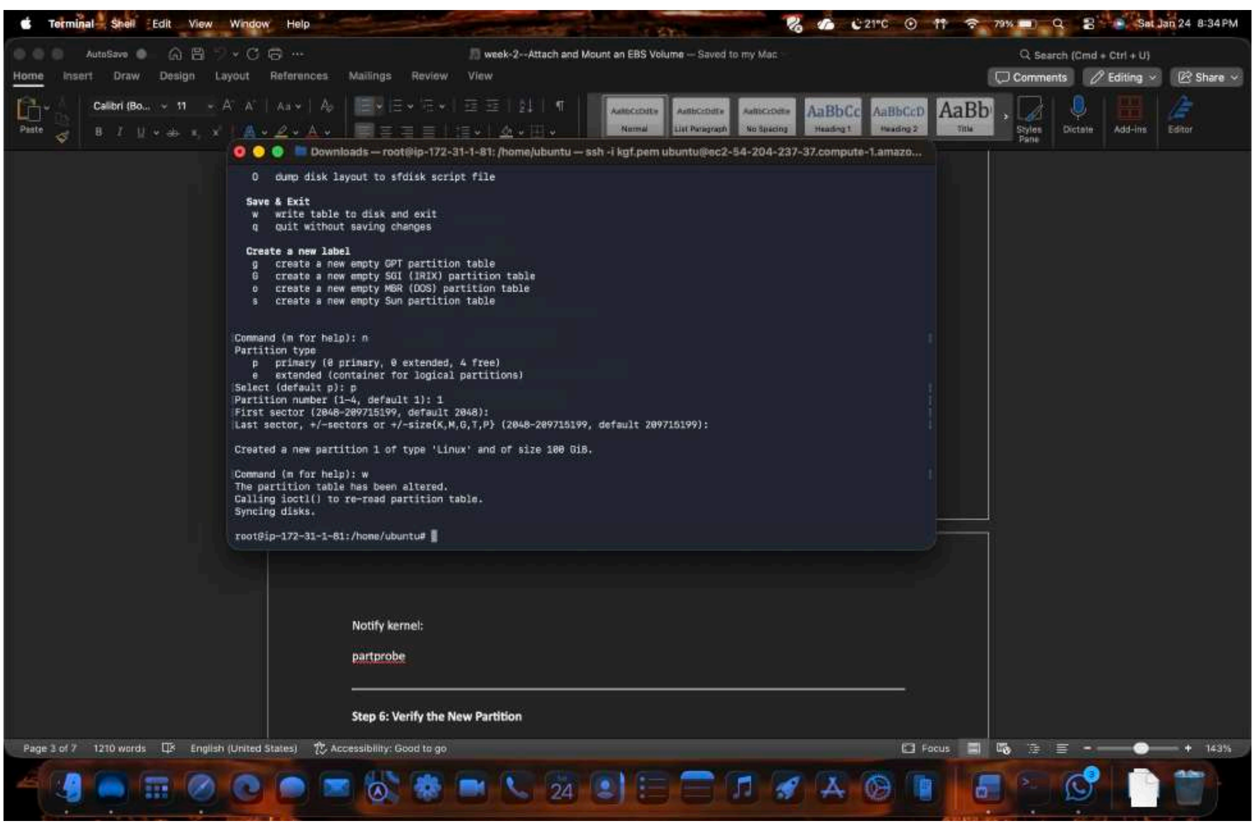
Create a partition on the new volume:

fdisk /dev/nme1n1

Inside fdisk:

- Press m → Help menu
- Press n → New partition
- Press p → Primary partition
- Press Enter → Default partition number
- Press Enter → Default first sector
- Press Enter → Default last sector (use full disk)
- Press w → Write changes and exit


Downloads — root@ip-172-31-1-81: /home/ubuntu — ssh -i kgf.pem ubuntu@ec2-54-204-237-37.compute-1.amazo...
ubuntu@ip-172-31-1-81:~$ sudo su
root@ip-172-31-1-81:/home/ubuntu# fdisk /dev/nvme1n1
Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.
Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x5c5ba0a2.
Command (m for help): m
Help:
DOS (MBR)
a toggle a bootable flag
b edit nested BSD disklabel
c toggle the dos compatibility flag
Generic
d delete a partition
F list free unpartitioned space
l list known partition types
n add a new partition
p print the partition table
t change a partition type
v verify the partition table
i print information about a partition
Misc
Press Enter → Default last sector (use full disk)
Press w → Write changes and exit

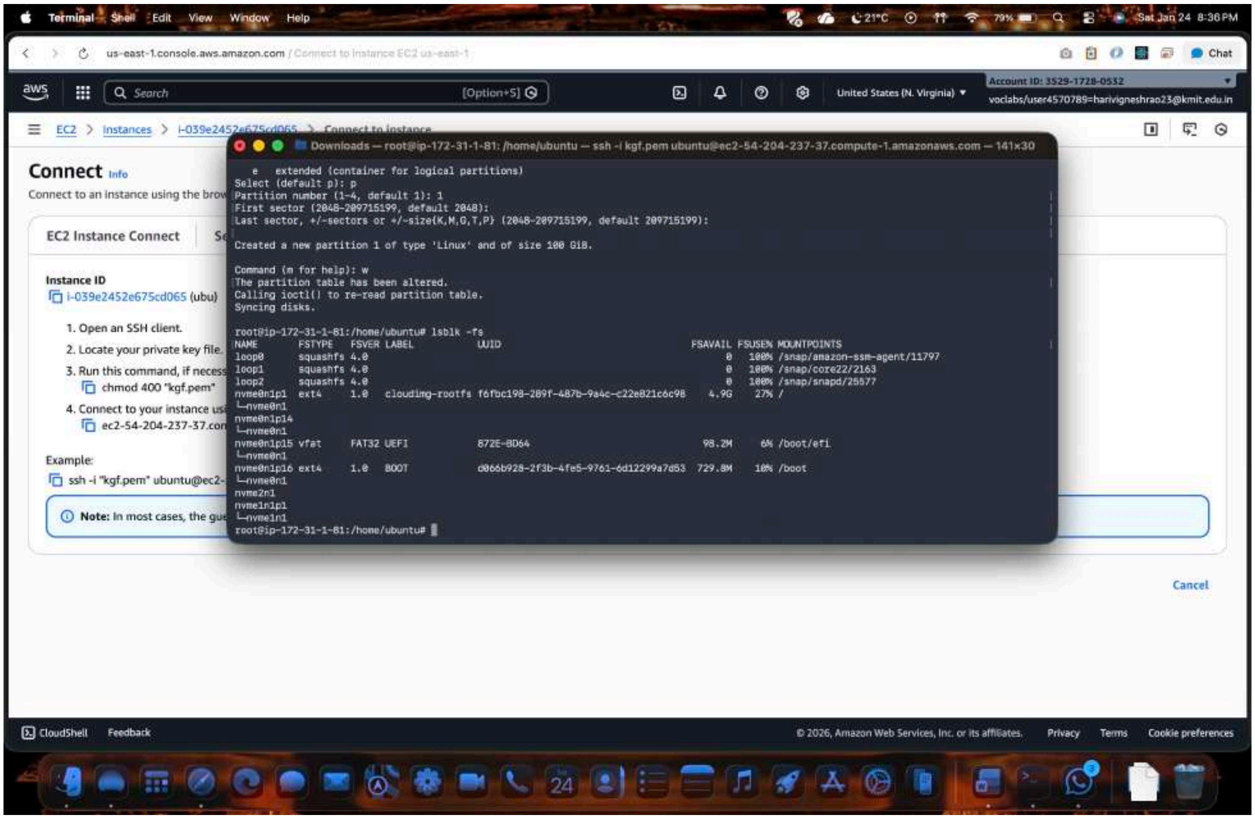

week-2--Attach and Mount an EBS Volume — Saved to my Mac
Downloads — root@ip-172-31-1-81: /home/ubuntu — ssh -i kgf.pem ubuntu@ec2-54-204-237-37.compute-1.amazo...
O dump disk layout to sfdisk script file
Save & Exit
w write table to disk and exit
q quit without saving changes
Create a new label
g create a new empty GPT partition table
G create a new empty SGI (IRIX) partition table
o create a new empty MBR (DOS) partition table
s create a new empty Sun partition table
Command (m for help): n
Partition type
p primary (0 primary, 0 extended, 4 free)
e extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-209715199, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-209715199, default 209715199):
Created a new partition 1 of type 'Linux' and of size 100 GiB.
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
root@ip-172-31-1-81:/home/ubuntu#
Notify kernel:
partprobe
Step 6: Verify the New Partition

Step 6: Verify the New Partition

lsblk -fs

You should now see:

/dev/xvdf1



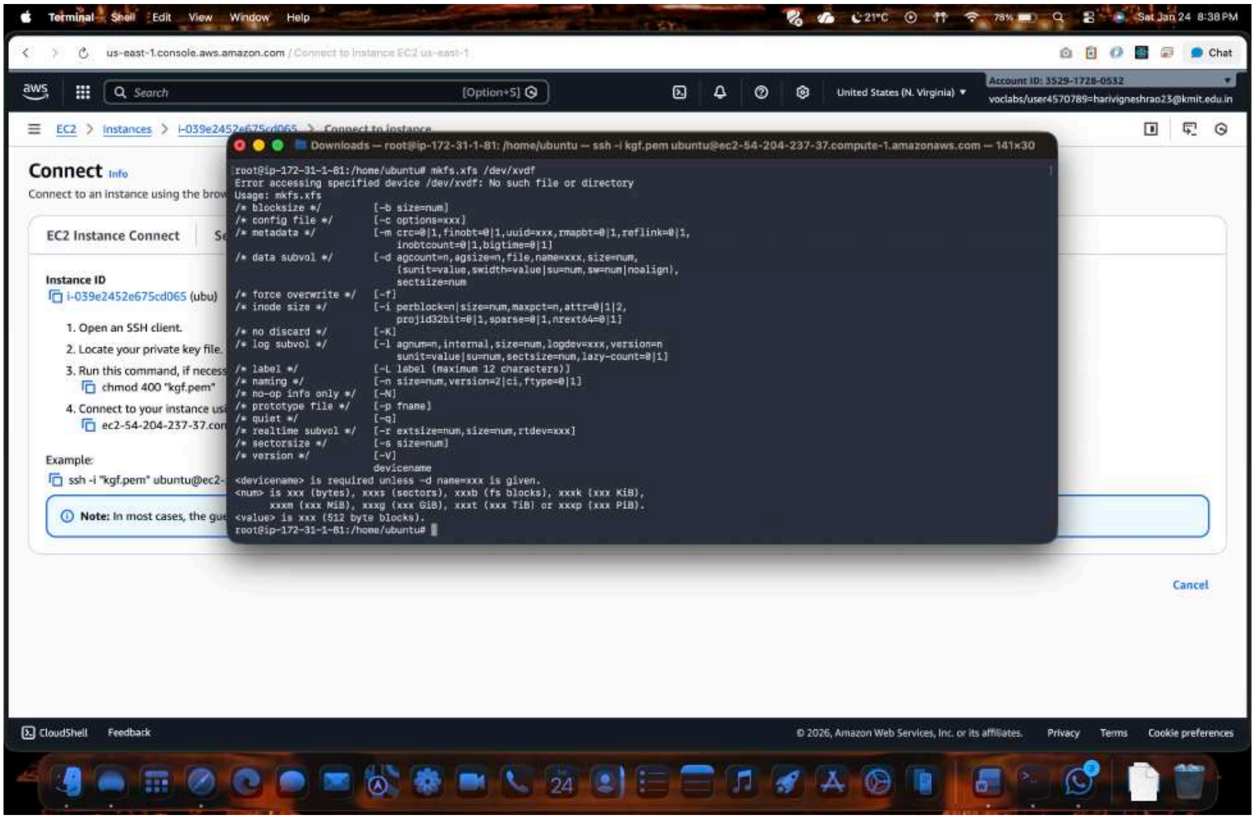
Step 7: Format the Partition

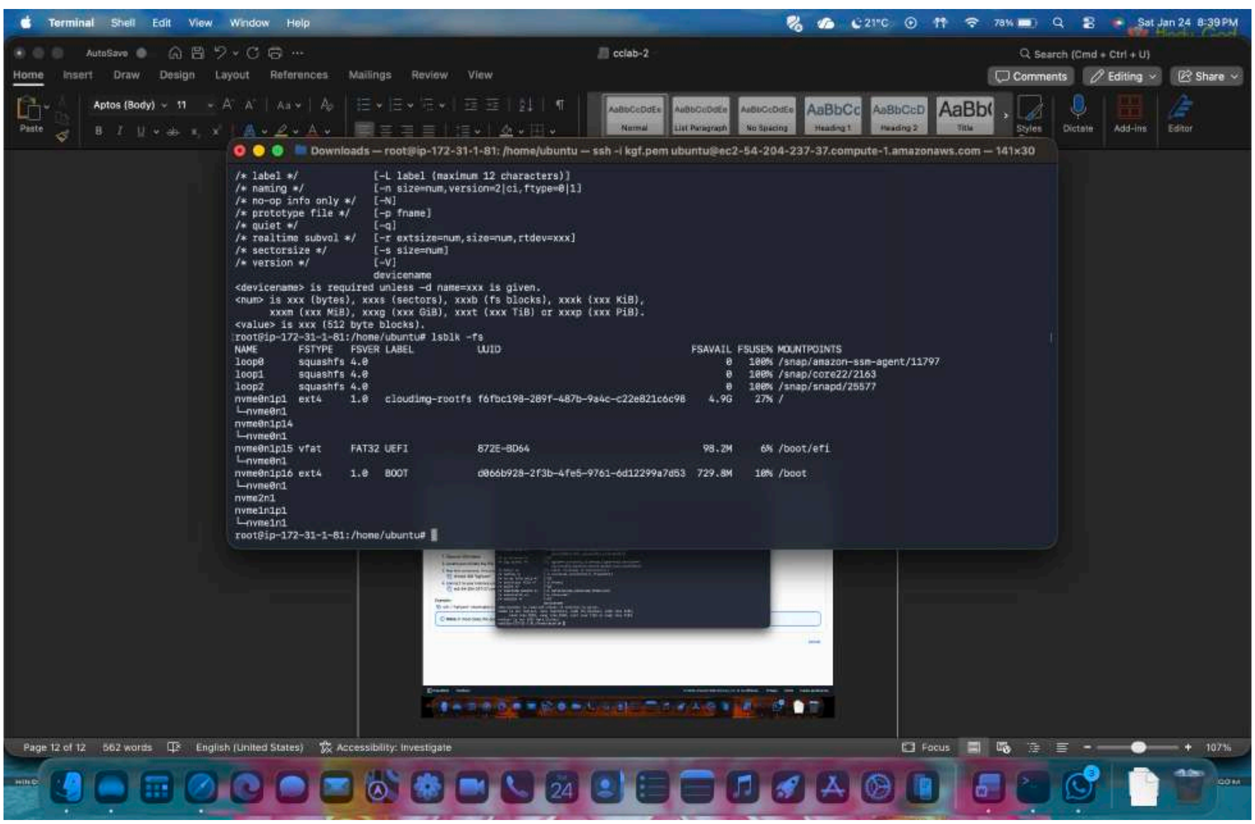
Format with XFS filesystem:

mkfs.xfs /dev/xvdf1

Verify:

lsblk -fs


Downloads — root@ip-172-31-1-81: /home/ubuntu — ssh -i kgf.pem ubuntu@ec2-54-204-237-37.compute-1.amazonaws.com — 141×30
root@ip-172-31-1-81:/home/ubuntu# mkfs.xfs /dev/xvdf
Error accessing specified device /dev/xvdf: No such file or directory
Usage: mkfs.xfs
/* blocksize */ [-b size=num]
/* config file */ [-c options=xxx]
/* metadata */ [-m crc=0|1,finobt=0|1,uuid=xxx,rmapbt=0|1,reflink=0|1, inobtcount=0|1,bigtime=0|1]
/* data subvol */ [-d agcount=n,agsize=n,file,name=xxx,size=num, (sunit=value,swidth=value|su=num,sw=num|noalign), sectsize=num
/* force overwrite */ [-f]
/* inode size */ [-i perblock=n|size=num,maxpct=n,attr=0|1|2, projid32bit=0|1,sparse=0|1,nrext64=0|1]
/* no discard */ [-K]
/* log subvol */ [-l agnum=n,internal,size=num,logdev=xxx,version=n sunit=value|su=num,sectsize=num,lazy-count=0|1]
/* label */ [-L label (maximum 12 characters)]
/* naming */ [-n size=num,version=2|ci,ftype=0|1]
/* no-op info only */ [-N]
/* prototype file */ [-p fname]
/* quiet */ [-q]
/* realtime subvol */ [-r extsize=num,size=num,rtdev=xxx]
/* sectorsize */ [-s size=num]
/* version */ [-V]
devicename
<devicename> is required unless -d name=xxx is given.
<num> is xxx (bytes), xxxs (sectors), xxxb (fs blocks), xxxk (xxx KiB), xxxm (xxx MiB), xxxg (xxx GiB), xxxt (xxx TiB) or xxxp (xxx PiB).
<value> is xxx (512 byte blocks).
root@ip-172-31-1-81:/home/ubuntu#


Downloads — root@ip-172-31-1-81: /home/ubuntu — ssh -i kgf.pem ubuntu@ec2-54-204-237-37.compute-1.amazonaws.com — 141×30
/* label */ [-L label (maximum 12 characters)]
/* naming */ [-n size=num,version=2|ci,ftype=0|1]
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/* sectorsize */ [-s size=num]
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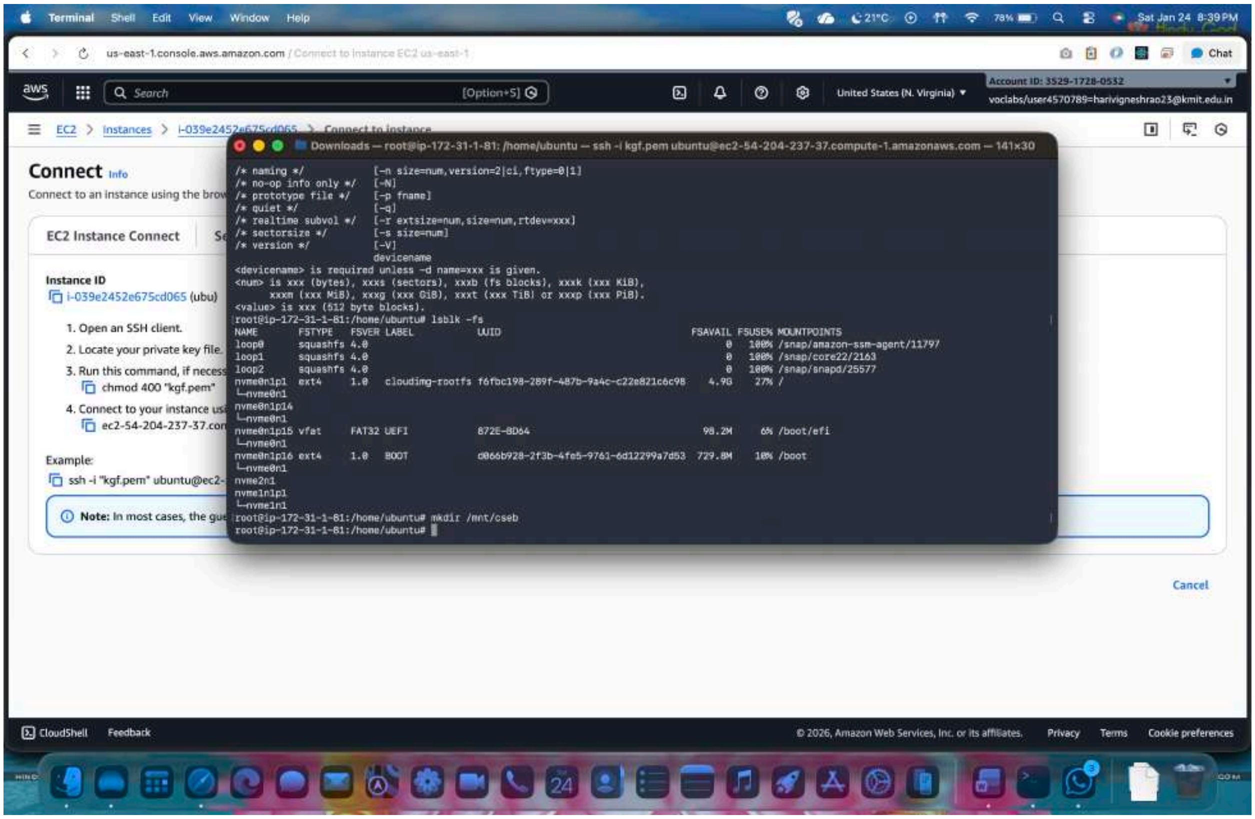
<value> is xxx (512 byte blocks).
root@ip-172-31-1-81:/home/ubuntu# lsblk -fs
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE% MOUNTPOINTS
loop0 squashfs 4.0 0 100% /snap/amazon-ssm-agent/11797
loop1 squashfs 4.0 0 100% /snap/core22/2163
loop2 squashfs 4.0 0 100% /snap/snapd/25577
nvme0n1p1 ext4 1.0 cloudimg-rootfs f6fbc198-289f-487b-9a4c-c22e821c6c98 4.9G 27% /
└─nvme0n1
nvme0n1p14
└─nvme0n1
nvme0n1p15 vfat FAT32 UEFI 872E-8D64 98.2M 6% /boot/efi
└─nvme0n1
nvme0n1p16 ext4 1.0 BOOT d066b928-2f3b-4fe5-9761-6d12299a7d53 729.8M 10% /boot
└─nvme0n1
nvme2n1
nvme1n1p1
└─nvme1n1
root@ip-172-31-1-81:/home/ubuntu#

Step 8: Create a Mount Directory

mkdir /mnt/cseb

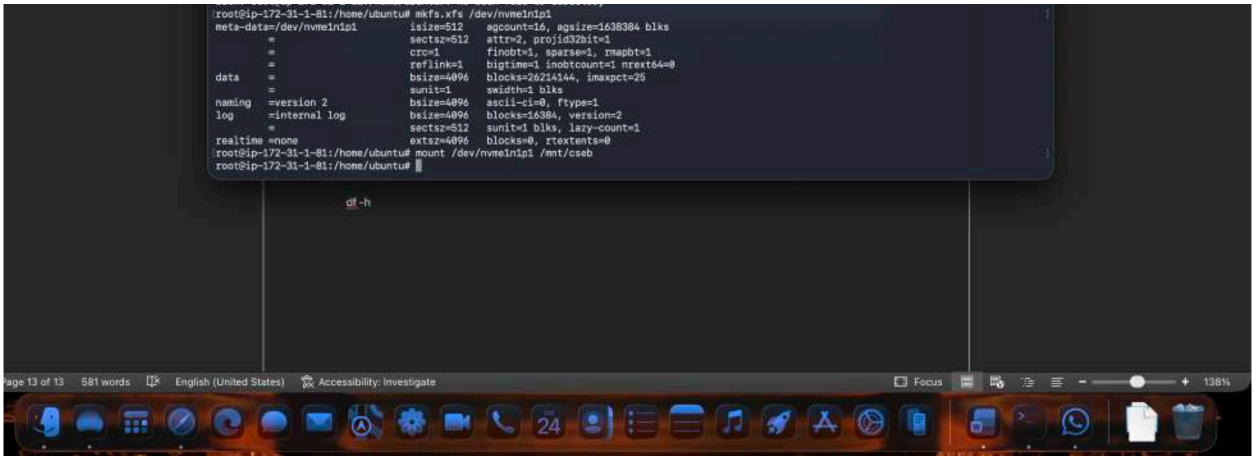


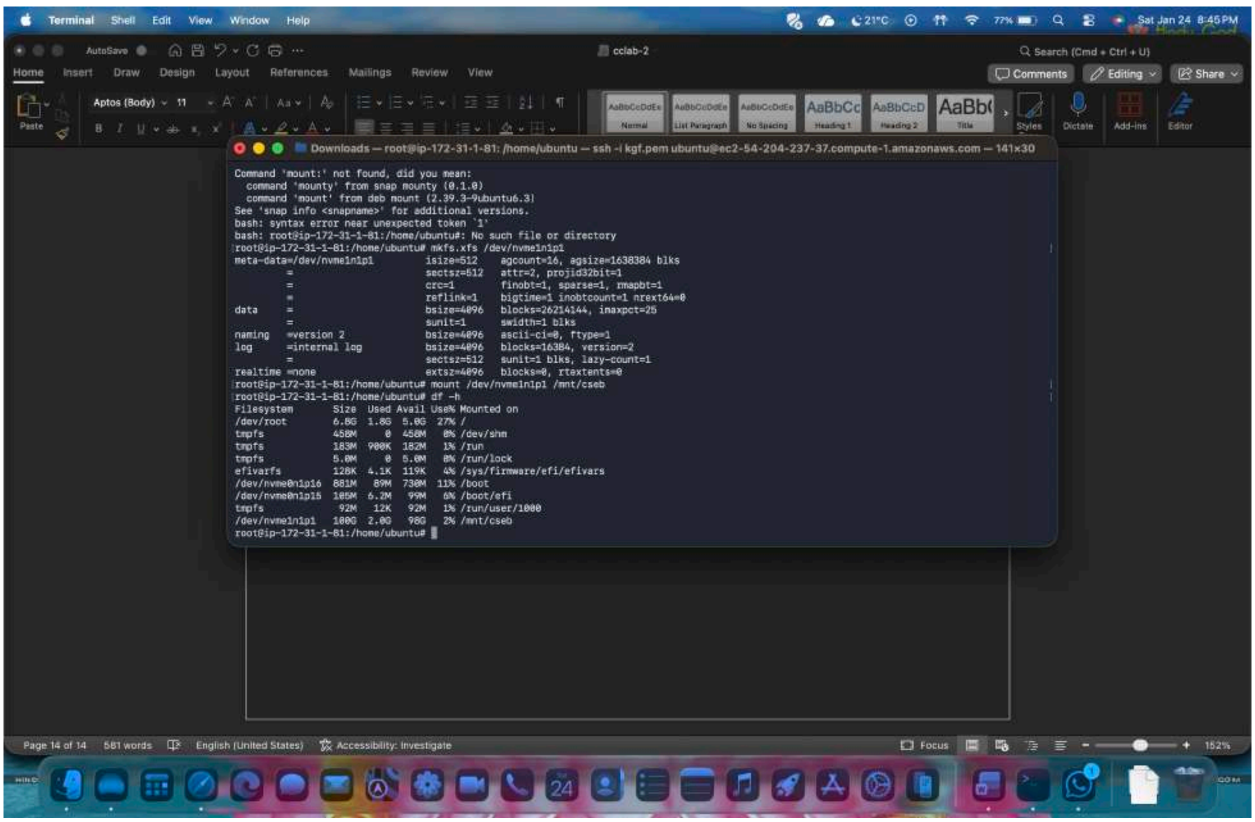
Step 9: Mount the Volume

mount /dev/xvdf1 /mnt/savitha

Verify:

df -h





Step 10: Make the Mount Persistent

Get the UUID of the volume:

blkid /dev/xvdf1

Edit fstab:

nano /etc/fstab

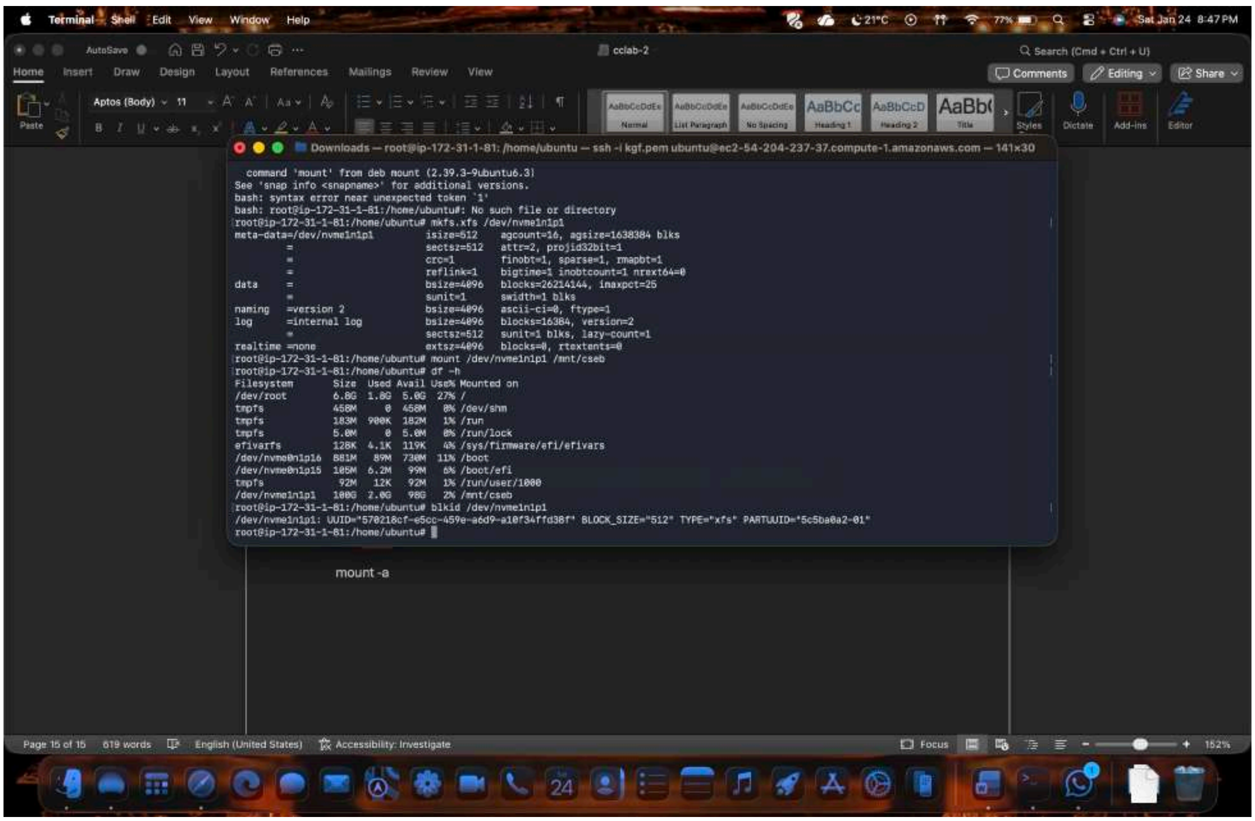
Add this line at the end:

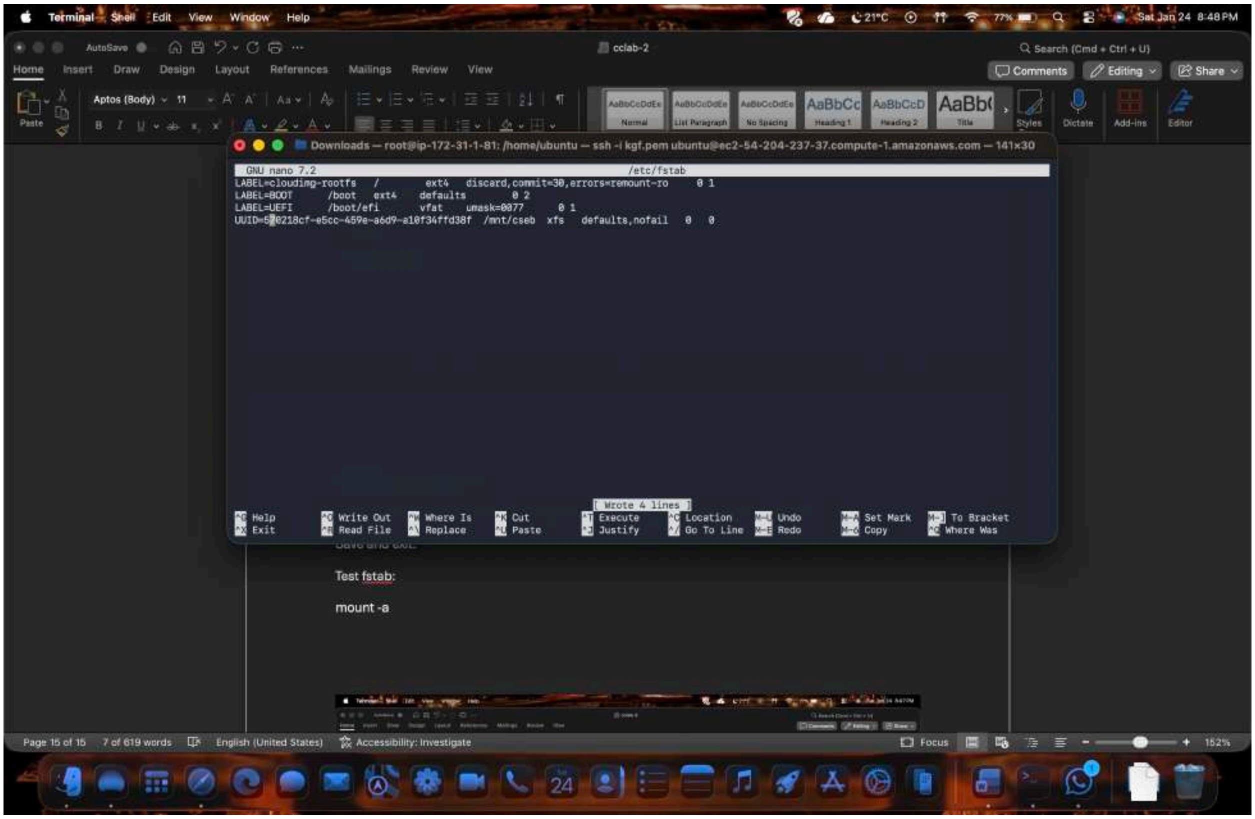
UUID= 8a979da3-1950-4810-bc0a-62c7a039d5cf /mnt/cseb xfs defaults,nofail 0 0

Save and exit.

Test fstab:

mount -a





Step 11: Verify Persistence

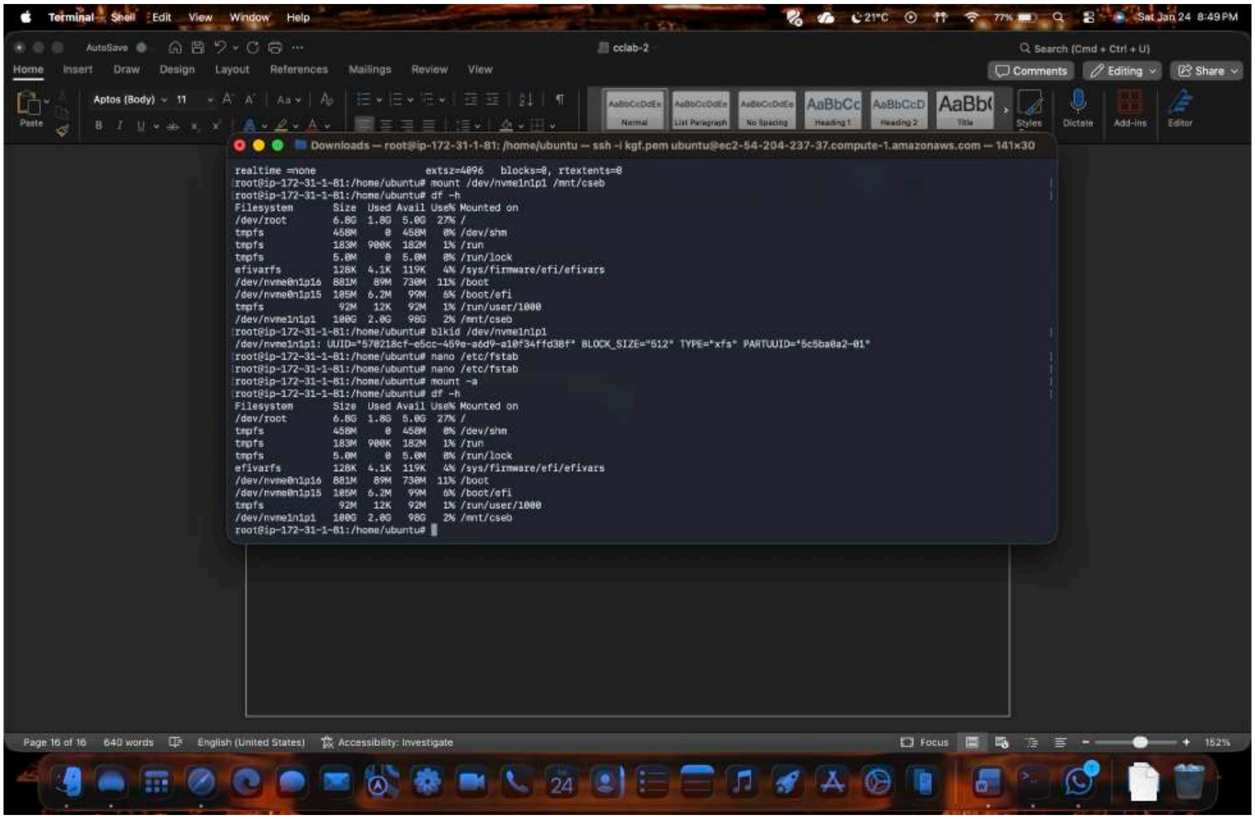
1. Reboot the instance:

reboot

2. Reconnect and check:

df -h

➡️ Volume should be mounted automatically.



Step 12: Detach and Reattach Volume (Data Recovery Scenario)

Detach:

1. EC2 → Volumes → Select volume
2. Actions → Detach volume

Attach to another EC2 instance:

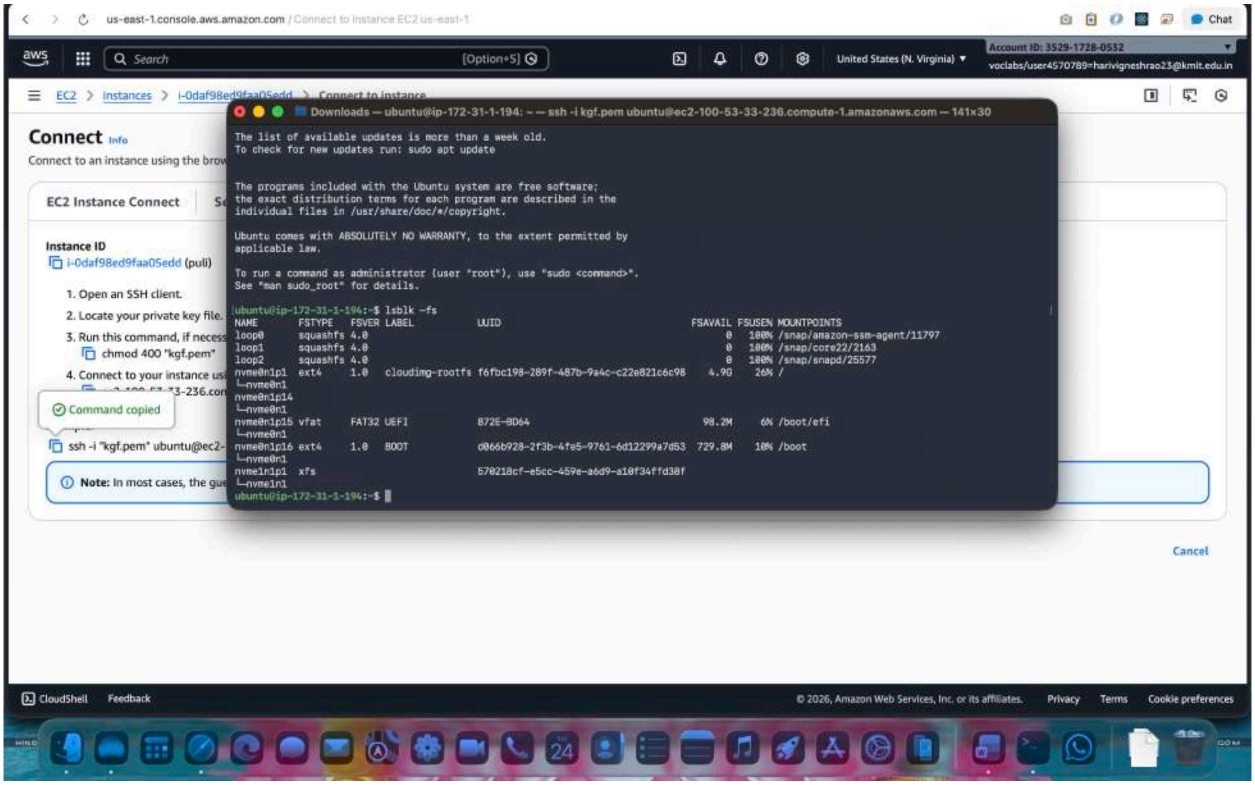
1. Attach to new instance (same AZ)
2. Login to new instance
3. Create mount directory:

mkdir /mnt/archana

4. Mount the volume:

mount /dev/xvdf1 /mnt/archana

All previous data will be available



Summary Flow

Create Volume → Attach → lsblk → fdisk → mkfs → mkdir → mount → fstab → reboot

3. Create a snapshot of the attached EBS volume and use it to create and attach a new volume to an EC2 instance in another AWS region.

Attaching Volumes across regions of EC2 using the snapshot

To create a snapshot of an EBS volume and attach it to an instance in another region in AWS, you'll need to follow these steps:

Step 1. Create a Snapshot:

a. Sign in to the AWS Management Console.

b. Navigate to the EC2 Dashboard.

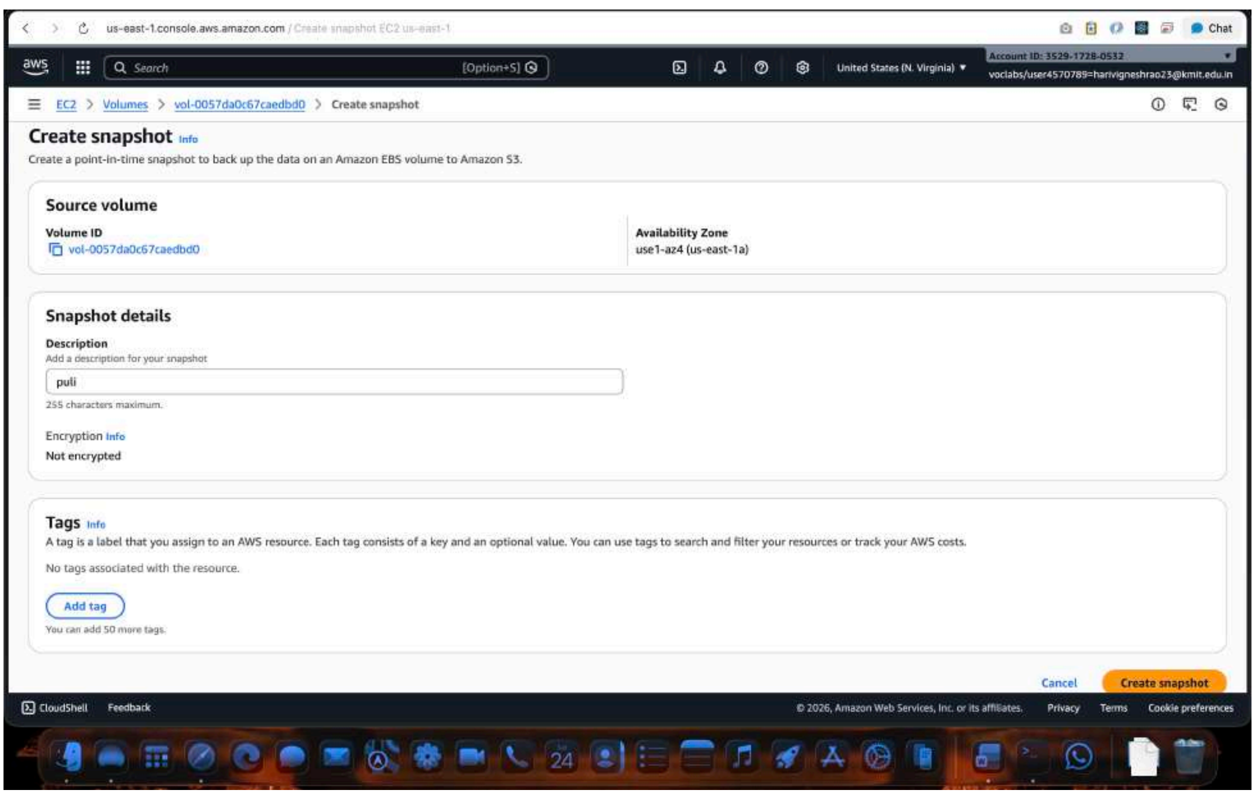
c. Click on "Volumes" in the left-hand navigation pane.

d. Select the EBS volume you want to create a snapshot of.

e. Click on the "Actions" dropdown menu above the volume list and select "Create Snapshot."

f. Provide a name and description for the snapshot.

g. Click on the "Create Snapshot" button to initiate the snapshot creation process.



Step 2. Copy the Snapshot to Another Region:

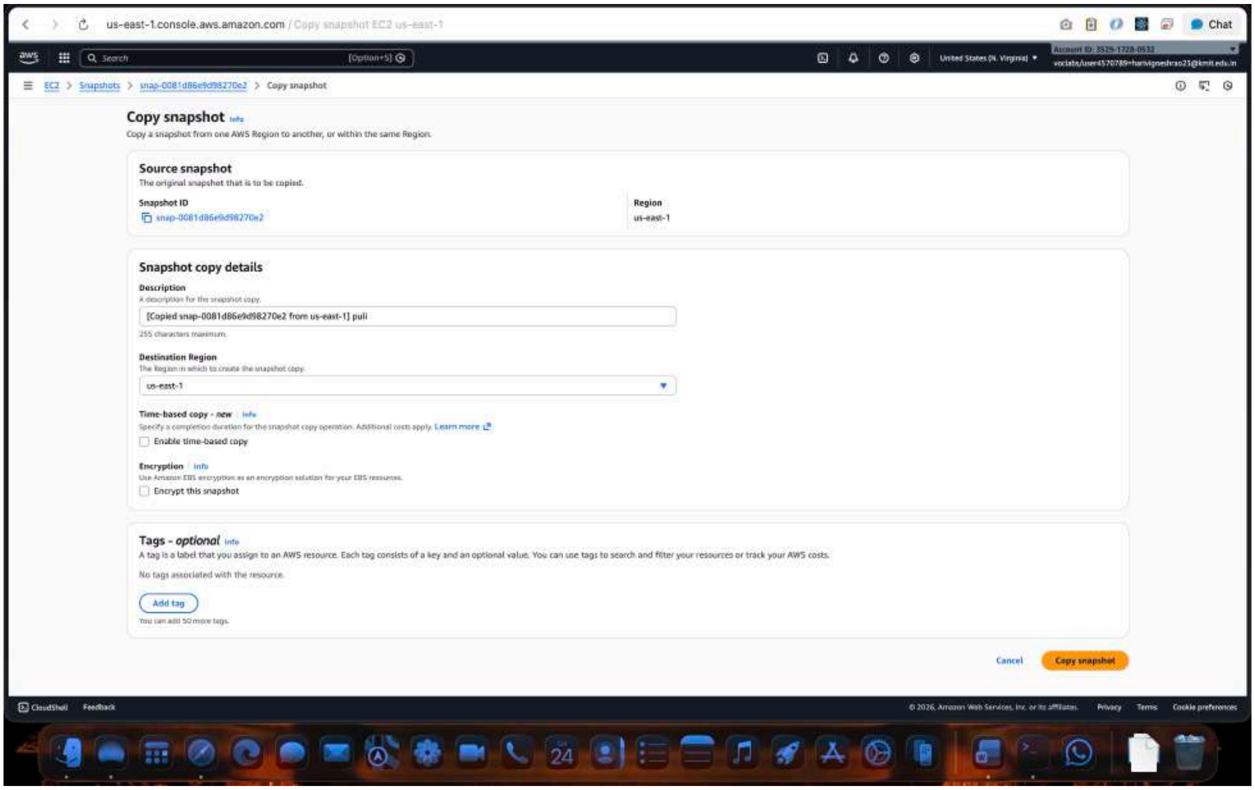
a. Once the snapshot is created, go to the "Snapshots" section in the EC2 Dashboard.

b. Select the snapshot you just created.

c. Click on the "Actions" dropdown menu above the snapshot list and select "Copy Snapshot."

d. Choose the destination region where you want to copy the snapshot.

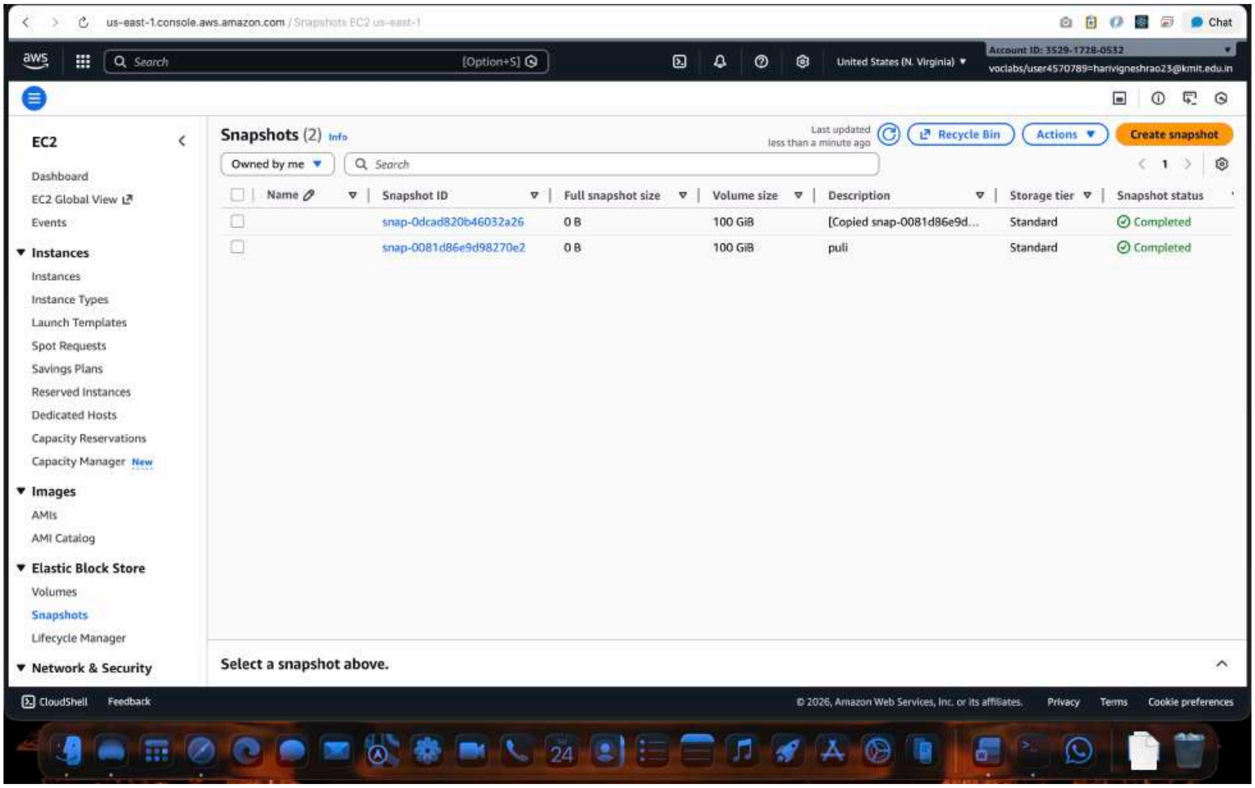
e. Click on the "Copy Snapshot" button to initiate the copy process. This may take some time depending on the size of the snapshot and the network speed.



Step 3. Monitor the Snapshot Copy Progress:

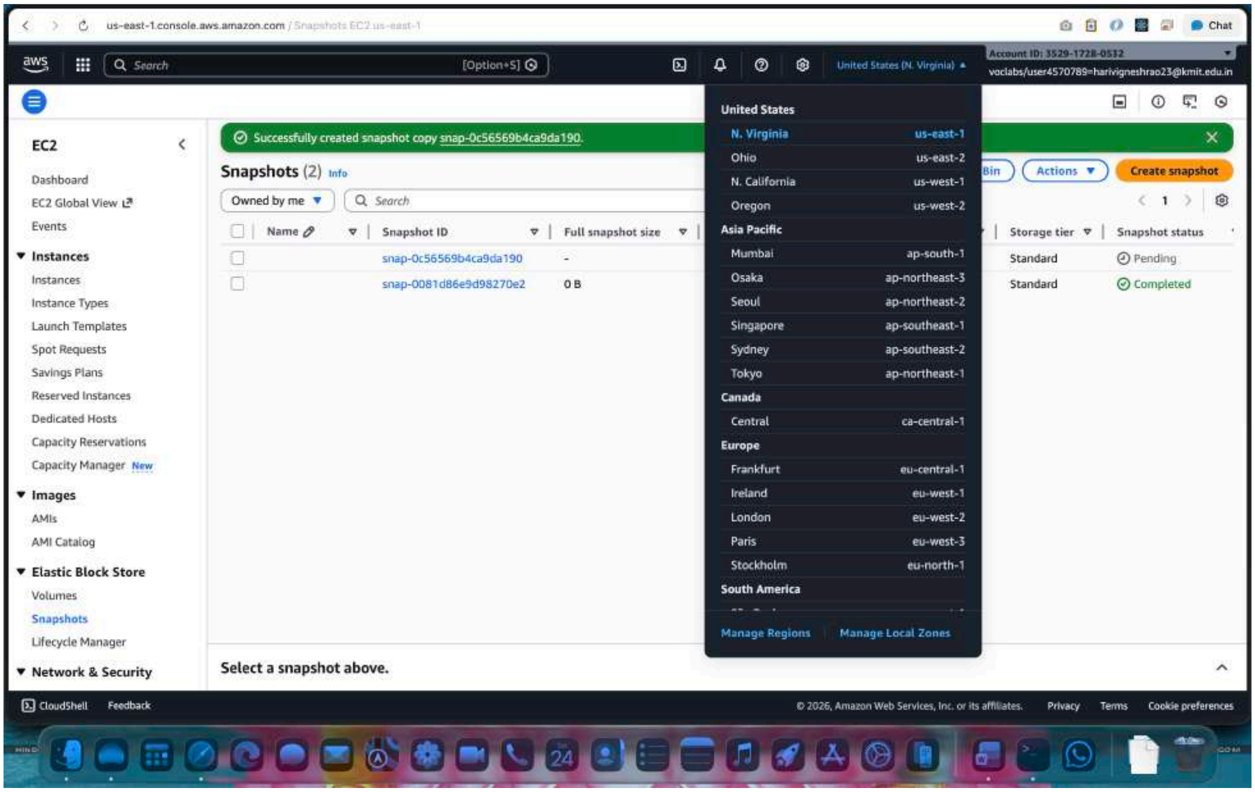
a. You can monitor the progress of the snapshot copy by navigating to the "Snapshots" section in the EC2 Dashboard of the source region.

b. Look for the snapshot you copied and check its status. It will change from "pending" to "completed" once the copy process is finished.



Step 4. Switch to the Destination Region:

a. Use the region selector in the top-right corner of the AWS Management Console to switch to the destination region where you copied the snapshot.



Step 5. Create a Volume from the Snapshot:

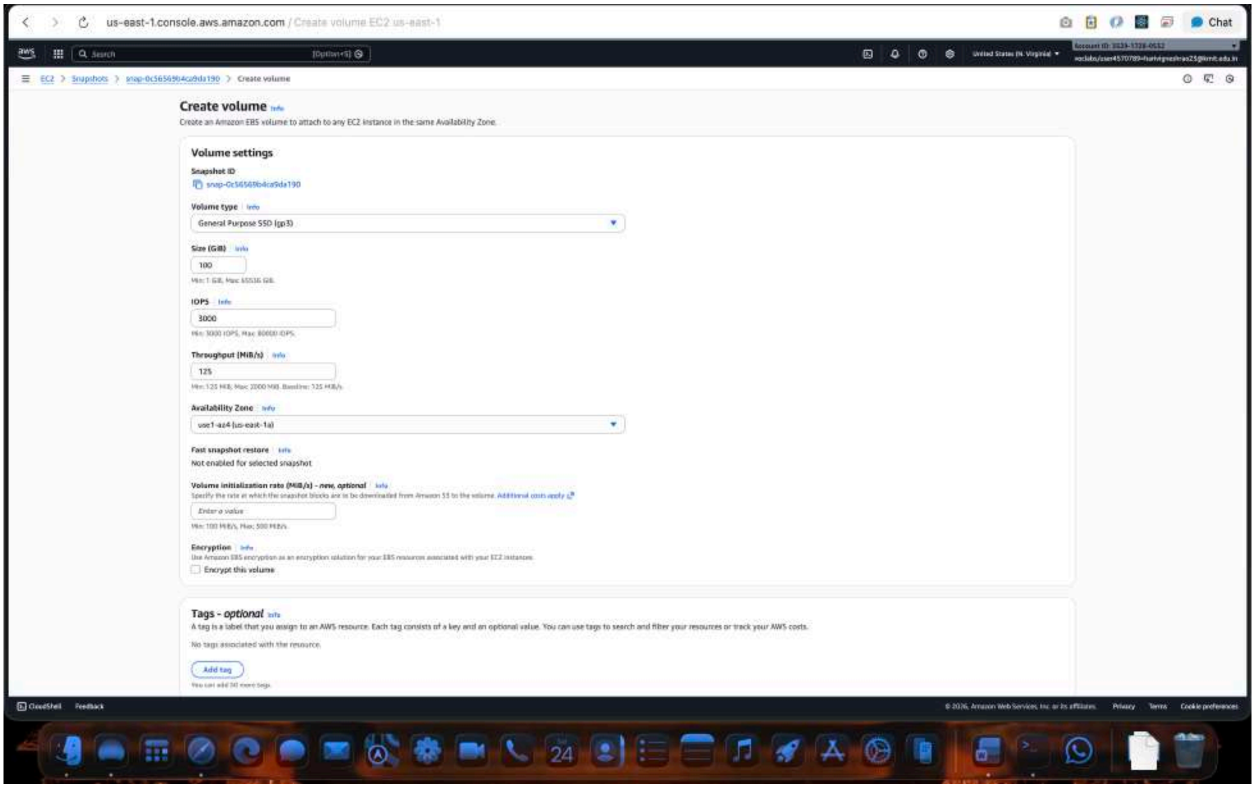
a. In the EC2 Dashboard of the destination region, go to the "Snapshots" section.

b. Find the snapshot you copied from the source region.

c. Click on the snapshot, then click on the "Actions" dropdown menu and select "Create Volume."

d. Configure the volume settings, such as volume type, size, and availability zone.

e. Click on the "Create Volume" button to create the volume from the snapshot.



Step 6. Create the ec2 instance & Attach the Volume to an Instance:

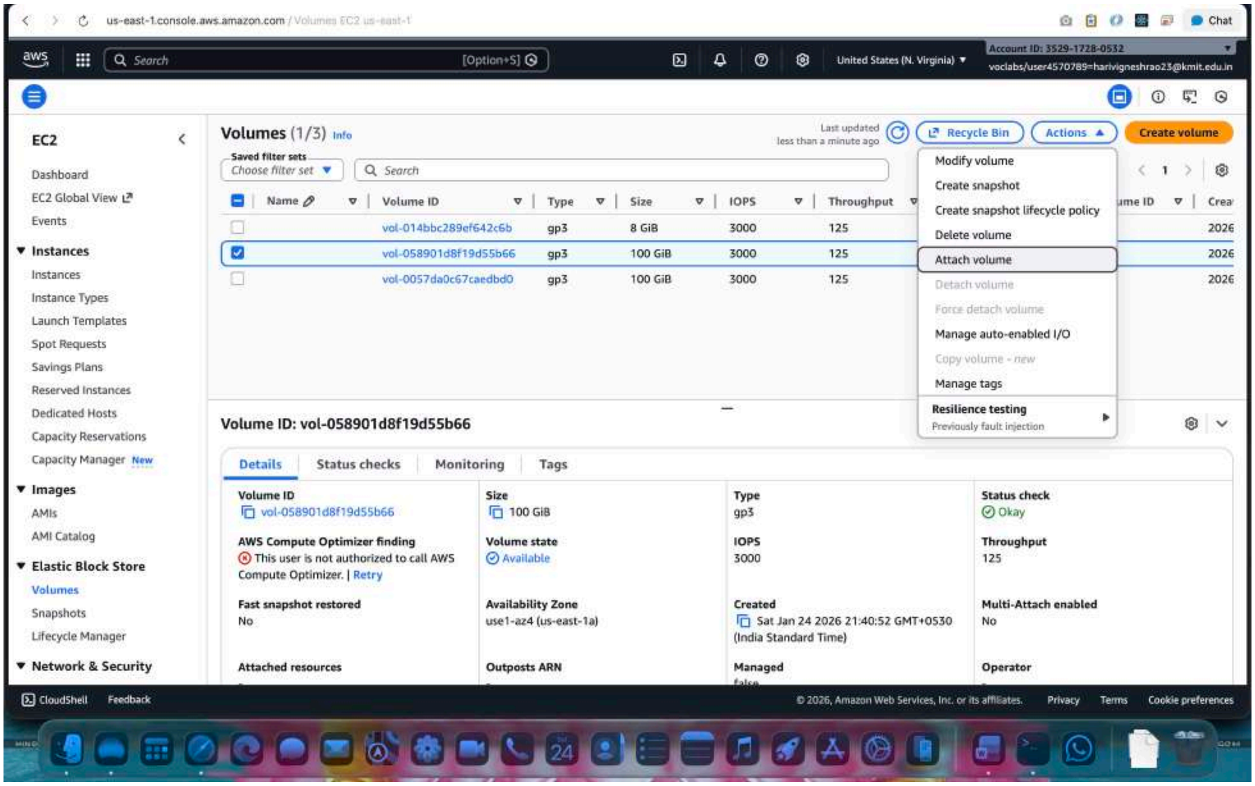
a. Once the volume is created, navigate to the "Volumes" section in the EC2 Dashboard.

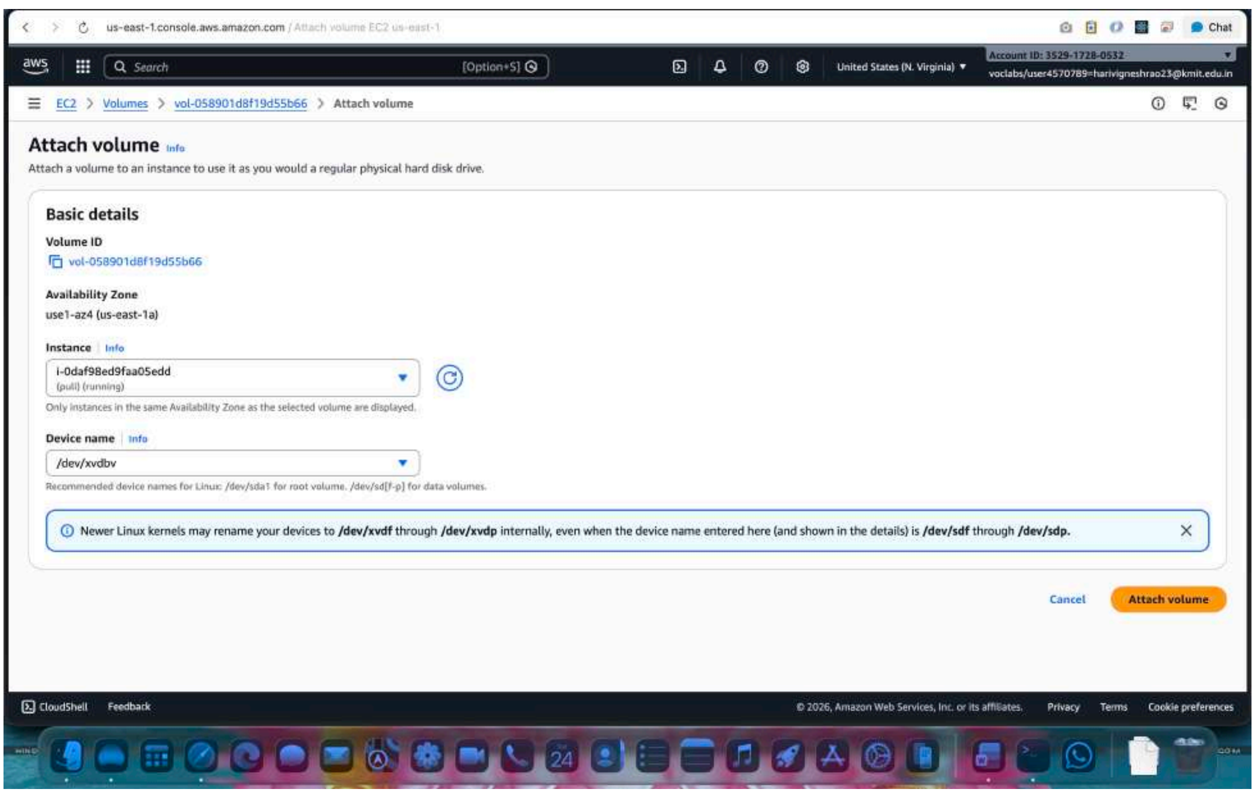
b. Find the newly created volume and select it.

c. Click on the "Actions" dropdown menu and select "Attach Volume."

D the EC2 instance to which you want to attach the volume and specify the device name.

e. Click on the "Attach" button to attach the volume to the instance.





We have created a snapshot of an EBS volume, copied it to another region, created a volume from the snapshot, and attached it to an instance in the destination region.